

Lab 12 Section 4: Population Scale Analysis

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11 May 2026

Q13

```
# Read the data
expr <- read.table("rs8067378_ENSG00000172057.6.txt", header = TRUE)

# Sample size for each genotype
table(expr$geno)
```

```
##
## A/A A/G G/G
## 108 233 121
```

```
# Median expression for each genotype
tapply(expr$exp, expr$geno, median)
```

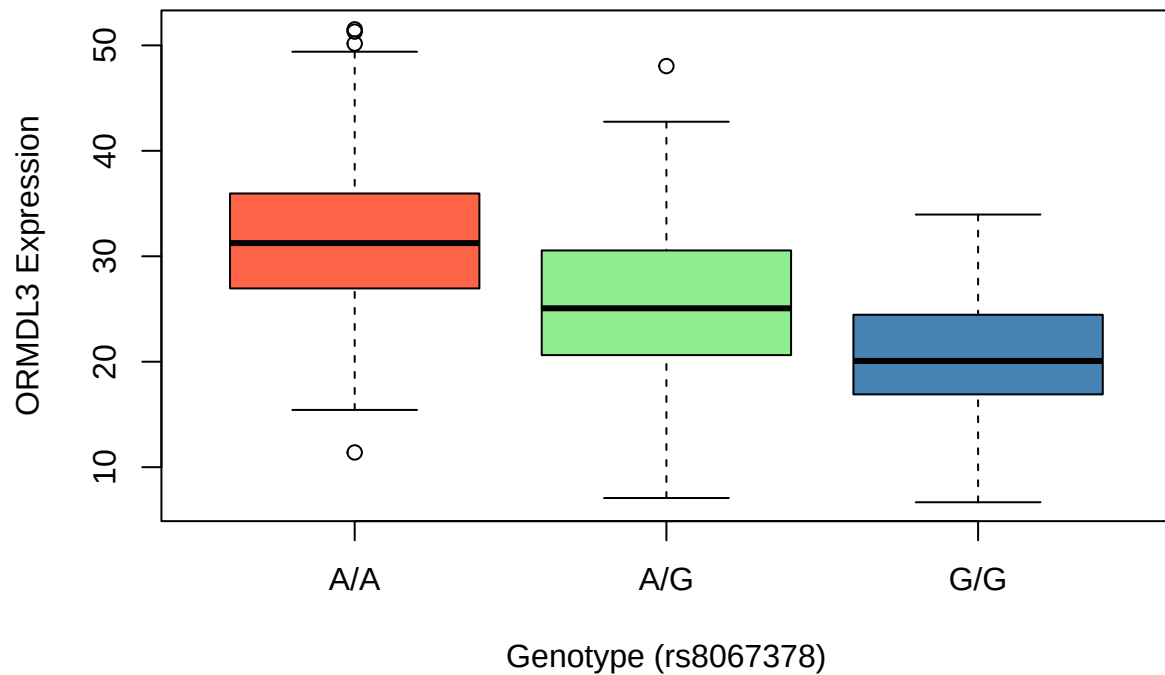
```
##      A/A      A/G      G/G
## 31.24847 25.06486 20.07363
```

The sample sizes are: A/A (n=108), A/G (n=233), and G/G (n=121). The corresponding median expression levels are: A/A = 31.25, A/G = 25.07, and G/G = 20.07. There is a clear decrease in median expression with each additional copy of the G allele.

Q14

```
# Boxplot of expression by genotype
boxplot(exp ~ geno, data = expr,
        xlab = "Genotype (rs8067378)",
        ylab = "ORMDL3 Expression",
        main = "Expression by Genotype for SNP rs8067378",
        col = c("tomato", "lightgreen", "steelblue"))
```

Expression by Genotype for SNP rs8067378



From the boxplot, A/A individuals have the highest ORMDL3 expression, A/G is intermediate, and G/G has the lowest. This indicates that the SNP rs8067378 does affect ORMDL3 expression — each copy of the G allele is associated with roughly 5-6 units lower expression. The asthma-risk G allele therefore appears to reduce ORMDL3 transcription in an additive manner.